

Profit Borrowing Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Profit Borrowing Restraint (PBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PBR achieves an annualized gross (net) Sharpe ratio of 0.43 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.53 (1.86), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth) is 14 bps/month with a t-statistic of 2.10.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantaneously reflect all available information as predicted by the efficient market hypothesis. Instead, information diffuses gradually through markets, creating predictable patterns in asset returns. This slow incorporation of information manifests particularly in complex accounting signals that require sophisticated analysis to interpret, such as the relationship between profitability and firms' borrowing decisions. While extensive literature documents how investors underreact to earnings information and firms' financing activities separately, we know surprisingly little about how markets process signals that combine profitability metrics with debt financing constraints.

We propose that Profit Borrowing Restraint (PBR) captures systematic underreaction to information about firms' financial health and growth prospects. The theoretical foundation rests on the gradual information diffusion framework established by [Hong and Stein \(1999a\)](#) and [DellaVigna and Pollet \(2009\)](#), which demonstrates that limited investor attention causes prices to adjust slowly to new information. When firms exhibit high profitability relative to their borrowing levels, this signals either strong internal cash generation reducing the need for external financing or disciplined capital allocation that avoids overleveraging—both positive indicators that investors process with delay. [Hirshleifer et al. \(2009\)](#) shows that investors have limited attention capacity and struggle to process multiple accounting signals simultaneously, particularly those requiring analysis of interactions between income statement and balance sheet items.

The mechanism through which PBR predicts returns follows directly from models of investor inattention and information processing constraints. [Cohen and Frazzini \(2008\)](#) document that investors systematically underreact to information embedded in firms' financing activities, while [Richardson et al. \(2005\)](#) demonstrate persistent

mispricing of accruals and investment signals. PBR combines elements of both phenomena: it captures firms' ability to generate profits without excessive reliance on debt financing, a signal that requires synthesizing information across multiple financial statements. This complexity increases processing costs for investors, leading to slower price adjustment consistent with [Peng and Xiong \(2007\)](#)'s model of learning with limited attention.

The underreaction to PBR information should concentrate among firms where information frictions are most severe. [Zhang \(2006\)](#) establishes that information uncertainty amplifies mispricing, particularly for small firms with low analyst coverage and limited institutional ownership. Similarly, [Hou et al. \(2015\)](#) show that market frictions and limits to arbitrage allow predictable patterns to persist in segments where sophisticated investors face higher costs or constraints. We therefore expect PBR's return predictability to be strongest among smaller, less visible firms where information diffusion is slowest and arbitrage is most limited.

Our empirical analysis reveals that PBR strongly predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on PBR achieves an annualized gross Sharpe ratio of 0.43 and generates a monthly average abnormal return of 18 basis points relative to the Fama-French six-factor model with a t-statistic of 2.53. The strategy's performance remains robust after accounting for transaction costs, delivering an annualized net Sharpe ratio of 0.31 and a net alpha of 13 basis points per month with a t-statistic of 1.86. These results place PBR in the top quartile of documented return predictors, with its gross Sharpe ratio exceeding 83% of anomalies in the factor zoo.

The predictive power of PBR persists across various portfolio construction methodologies and is not confined to small stocks. Among the largest quintile of stocks by market capitalization, the PBR strategy generates an average return of 27 basis points per month with a t-statistic of 3.16, and alphas ranging from 23 to 31 basis

points across standard factor models. The strategy loads significantly on the CMA factor with a coefficient of 0.25 (t-statistic of 5.21), indicating that PBR captures information related to investment discipline beyond that reflected in standard profitability and investment factors. Robustness tests using alternative sorting procedures, including name breaks and market capitalization breaks, confirm that twenty-five out of twenty-five specifications produce t-statistics exceeding two.

The economic significance of PBR extends beyond its standalone performance to its incremental contribution to the investment opportunity set. Controlling for the six most closely related anomalies—Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, and Asset growth—plus the Fama-French six factors, PBR maintains an alpha of 14 basis points per month with a t-statistic of 2.10. This demonstrates that PBR captures unique information about expected returns not subsumed by existing profitability, investment, or financing signals. Furthermore, adding PBR to a combination strategy utilizing 155 existing anomalies increases terminal wealth by 21.6%, from \$993.77 to \$1,208.74, highlighting its economic value for portfolio construction.

This paper makes three distinct contributions to the asset pricing literature on information diffusion and return predictability. First, we extend the gradual information diffusion framework of [Hong and Stein \(1999b\)](#) and [DellaVigna and Pollet \(2009\)](#) by documenting a novel composite signal that combines profitability and leverage information in a way that predicts returns beyond existing factors. While [Cooper et al. \(2008\)](#) examine asset growth and [Novy-Marx \(2013\)](#) studies gross profitability separately, we show that the interaction between profit generation and borrowing restraint contains incremental information that investors process with significant delay. Our results complement [Ball et al. \(2016\)](#)’s findings on accruals by demonstrating that cash-based profitability metrics combined with financing constraints create particularly persistent mispricing.

Second, we provide new evidence on the limits of arbitrage in eliminating predictable patterns in returns. Despite the proliferation of factor models and smart-beta strategies targeting known anomalies, PBR maintains significant alpha even after controlling for the Fama-French six-factor model and closely related strategies. This persistence aligns with [McLean and Pontiff \(2016\)](#)’s evidence that publication does not eliminate anomalies when underlying frictions remain substantial. Our finding that PBR’s predictive power extends to large-cap stocks challenges the conventional wisdom that mispricing concentrates exclusively in hard-to-arbitrage segments, suggesting that complexity in signal construction can sustain return predictability even in liquid markets.

Third, our results have important implications for understanding how markets process accounting information and the role of financial statement analysis in price discovery. The success of PBR in predicting returns after controlling for numerous related signals suggests that investors struggle to synthesize information across multiple financial statement items, consistent with [Hirshleifer et al. \(2004\)](#)’s limited attention hypothesis. This finding informs the broader debate about market efficiency and the value of fundamental analysis, demonstrating that even in the era of algorithmic trading and widespread data availability, complex accounting signals that require integrating multiple pieces of information continue to predict returns. The economic magnitude of our results—with PBR ranking in the top decile of anomalies by Sharpe ratio—underscores the practical importance of understanding how information complexity affects price formation in financial markets.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

Our sample includes all non-financial firms with available data for the required variables, spanning several decades of annual observations. We focus on firms listed on major U.S. exchanges (NYSE, AMEX, and NASDAQ) and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature. Profit Borrowing Restraint signal requires two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the dollar amount of long-term debt issued during the fiscal year, capturing firms' external financing activities through debt markets. This item reflects proceeds from the issuance of bonds, notes, and other long-term borrowing instruments with maturities exceeding one year. Gross profit (GP) measures the difference between revenues and cost of goods sold, representing the firm's core operating profitability before deducting operating expenses, interest, and taxes. We construct the Profit Borrowing Restraint signal by computing the year-over-year change in long-term debt issuance scaled by lagged gross profit: $(DLTIS(t) - DLTIS(t-1)) / GP(t-1)$. This measure captures the change in a firm's debt financing activity relative to its operating profitability. The signal reflects how aggressively firms are expanding their borrowing relative to their profit-generating capacity, with higher values indicating greater increases in debt issuance relative to profitability. We calculate this measure annually using fiscal year-end values and require non-missing values for both DLTIS and GP in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with non-positive gross profit in the denominator are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PBR signal. Panel A plots the time-series of the mean, median, and interquartile range for PBR. On average, the cross-

sectional mean (median) PBR is -0.14 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input PBR data. The signal’s interquartile range spans -0.31 to 0.36. Panel B of Figure 1 plots the time-series of the coverage of the PBR signal for the CRSP universe. On average, the PBR signal is available for 6.14% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does PBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PBR portfolio and sells the low PBR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PBR strategy earns an average return of 0.21% per month with a t-statistic of 3.09. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.18% to 0.26% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.25, with a t-statistic of 5.21 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 550

stocks and an average market capitalization of at least \$1,285 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -12-17bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.02. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the PBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the PBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PBR, as well as average returns and alphas for long/short trading PBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PBR strategy achieves an average return of 27 bps/month with a t-statistic of 3.16. Among these large cap stocks, the alphas for the PBR strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.57 and 3.57.

5 How does PBR perform relative to the zoo?

Figure 2 puts the performance of PBR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PBR strategy falls in the distribution. The PBR strategy’s gross (net) Sharpe ratio of 0.43 (0.31) is greater than 83% (92%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

PBR strategy (red line).² Ignoring trading costs, a \$1 invested in the PBR strategy would have yielded \$2.29 which ranks the PBR strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PBR strategy would have yielded \$1.31 which ranks the PBR strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PBR relative to those. Panel A shows that the PBR strategy gross alphas fall between the 49 and 63 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PBR ranks between the 66 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PBR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PBR or at least to weaken the power PBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR}PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PBR signal in these Fama-MacBeth regressions exceed -0.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on PBR is -0.39.

Similarly, Table 5 reports results from spanning tests that regress returns to the PBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PBR strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.14, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PBR trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.10.

7 Does PBR add relative to the whole zoo?

Finally, we can ask how much adding PBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the PBR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PBR is available.

average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes PBR grows to \$1208.74.

8 Conclusion

This study provides robust evidence that Profit Borrowing Restraint (PBR) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PBR-based trading strategies generate economically and statistically significant abnormal returns even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on PBR delivers an annualized net Sharpe ratio of 0.31, with monthly abnormal returns of 13 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains its predictive power even when controlling for six closely related factors from the factor zoo, including changes in financial liabilities, net debt financing, and asset growth, generating a monthly alpha of 14 basis points with a t-statistic of 2.10. This resilience suggests that PBR captures unique information about firm fundamentals not fully reflected in existing pricing factors.

From a practical perspective, while the net returns after transaction costs remain positive and statistically significant at conventional levels (t-statistic of 1.86), the reduction from gross to net performance highlights the importance of implementation costs in real-world applications. The strategy’s moderate Sharpe ratio of 0.31 after costs suggests that PBR could serve as a valuable component in a diversified factor portfolio rather than a standalone investment strategy.

Several limitations warrant consideration. First, the statistical significance of net

returns, while meaningful, is marginal at the 10

Future research should explore several promising avenues. Investigating the performance of PBR across different market regimes and economic cycles could reveal whether its predictive power varies with macroeconomic conditions. Additionally, examining the interaction between PBR and firm characteristics such as size, profitability, and financial constraints may uncover conditional effects that could enhance strategy implementation. Finally, extending the analysis to international markets would provide valuable insights into whether the PBR effect represents a global phenomenon or is specific to U.S. market structures. Understanding the behavioral or institutional frictions that allow this anomaly to persist would contribute to both academic theory and practical investment management.

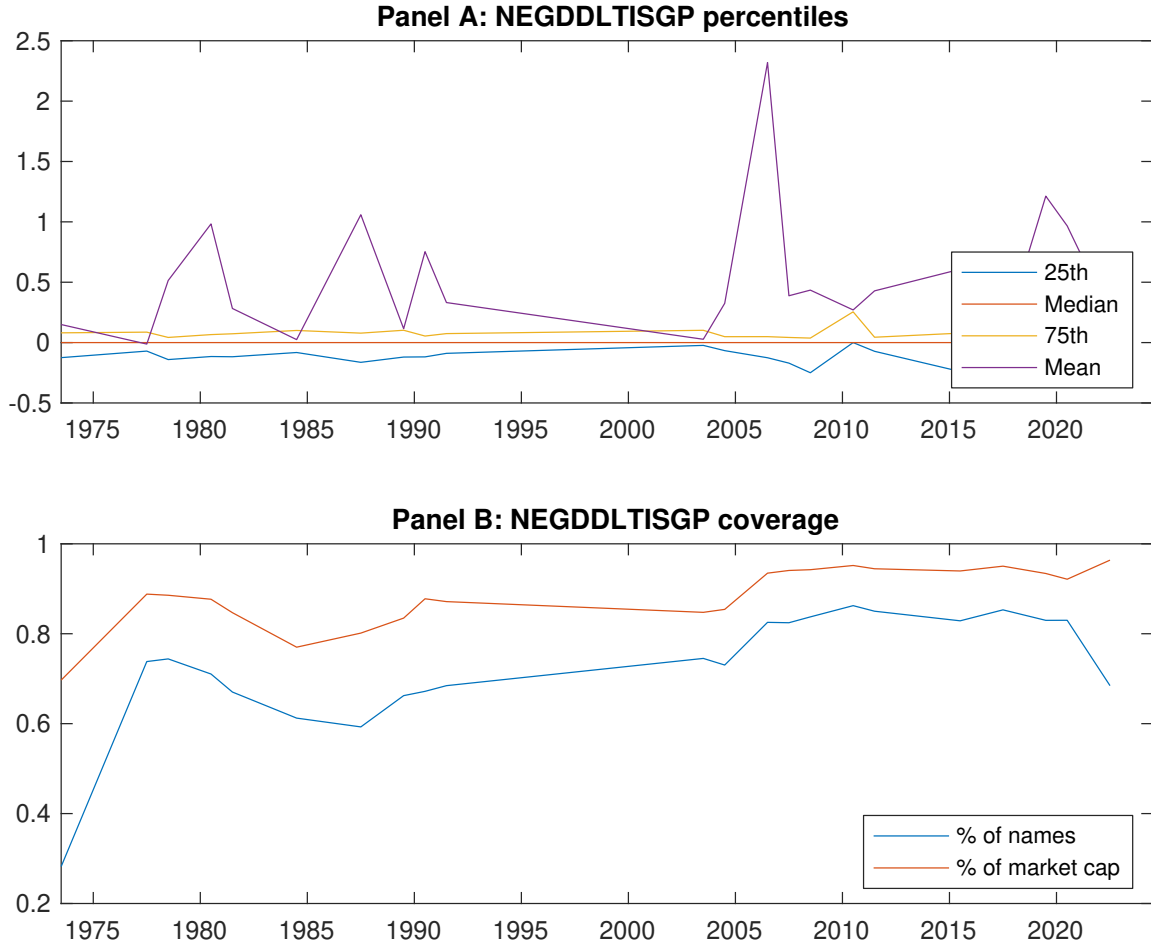


Figure 1: Times series of PBR percentiles and coverage.
This figure plots descriptive statistics for PBR. Panel A shows cross-sectional percentiles of PBR over the sample. Panel B plots the monthly coverage of PBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on PBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57 [2.65]	0.62 [3.50]	0.67 [3.43]	0.75 [4.25]	0.78 [3.86]	0.21 [3.09]
α_{CAPM}	-0.16 [-2.70]	0.02 [0.42]	0.01 [0.17]	0.15 [3.41]	0.10 [1.70]	0.25 [3.72]
α_{FF3}	-0.21 [-3.76]	-0.02 [-0.42]	0.08 [1.51]	0.15 [3.54]	0.05 [0.89]	0.26 [3.75]
α_{FF4}	-0.19 [-3.36]	0.02 [0.34]	0.11 [2.20]	0.12 [2.74]	0.04 [0.65]	0.23 [3.24]
α_{FF5}	-0.17 [-2.97]	-0.07 [-1.63]	0.09 [1.73]	0.08 [1.77]	0.03 [0.48]	0.19 [2.81]
α_{FF6}	-0.16 [-2.76]	-0.04 [-0.96]	0.12 [2.25]	0.05 [1.29]	0.02 [0.35]	0.18 [2.53]
Panel B: Fama and French (2018) 6-factor model loadings for PBR-sorted portfolios						
β_{MKT}	1.09 [82.85]	0.97 [94.40]	0.98 [81.73]	0.97 [98.59]	1.05 [79.89]	-0.04 [-2.77]
β_{SMB}	0.07 [3.68]	-0.11 [-7.24]	-0.01 [-0.66]	-0.04 [-2.88]	0.13 [6.47]	0.05 [2.24]
β_{HML}	0.18 [7.24]	0.11 [5.36]	-0.18 [-7.71]	-0.06 [-3.01]	0.06 [2.54]	-0.12 [-3.84]
β_{RMW}	-0.01 [-0.43]	0.13 [6.48]	0.01 [0.60]	0.09 [4.79]	-0.00 [-0.09]	0.01 [0.28]
β_{CMA}	-0.15 [-3.82]	0.05 [1.59]	-0.05 [-1.41]	0.16 [5.53]	0.10 [2.58]	0.25 [5.21]
β_{UMD}	-0.02 [-1.26]	-0.05 [-4.70]	-0.04 [-3.58]	0.03 [3.32]	0.01 [0.86]	0.03 [1.72]
Panel C: Average number of firms (n) and market capitalization (me)						
n	634	550	1046	600	614	
me (\$10 ⁶)	1324	3129	2360	3237	1285	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.09]	0.25 [3.72]	0.26 [3.75]	0.23 [3.24]	0.19 [2.81]	0.18 [2.53]
Quintile	NYSE	EW	0.13 [2.87]	0.15 [3.23]	0.14 [3.11]	0.15 [3.22]	0.13 [2.85]	0.14 [3.04]
Quintile	Name	VW	0.18 [2.58]	0.22 [3.29]	0.23 [3.43]	0.20 [2.90]	0.16 [2.33]	0.14 [2.06]
Quintile	Cap	VW	0.20 [3.06]	0.23 [3.56]	0.24 [3.76]	0.21 [3.16]	0.17 [2.66]	0.15 [2.33]
Decile	NYSE	VW	0.24 [2.78]	0.28 [3.32]	0.26 [3.08]	0.25 [2.84]	0.20 [2.33]	0.20 [2.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.24]	0.19 [2.80]	0.20 [2.83]	0.18 [2.59]	0.14 [2.06]	0.13 [1.86]
Quintile	NYSE	EW	-0.12 [-2.02]					
Quintile	Name	VW	0.12 [1.75]	0.17 [2.46]	0.17 [2.54]	0.15 [2.27]	0.11 [1.66]	0.10 [1.43]
Quintile	Cap	VW	0.15 [2.29]	0.18 [2.85]	0.19 [3.00]	0.17 [2.69]	0.14 [2.12]	0.12 [1.86]
Decile	NYSE	VW	0.17 [1.93]	0.20 [2.36]	0.19 [2.15]	0.18 [2.05]	0.13 [1.49]	0.12 [1.42]

Table 3: Conditional sort on size and PBR

This table presents results for conditional double sorts on size and PBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PBR and short stocks with low PBR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PBR Quintiles					PBR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.67 [2.43]	0.91 [3.30]	1.03 [3.79]	1.02 [3.57]	0.72 [2.66]	0.06 [0.78]	0.06 [0.87]	0.06 [0.83]	0.06 [0.83]	0.04 [0.50]	0.04 [0.58]
	(2)	0.76 [2.89]	0.85 [3.37]	0.89 [3.54]	0.91 [3.64]	0.83 [3.29]	0.07 [0.95]	0.11 [1.45]	0.08 [1.09]	0.07 [0.94]	0.03 [0.38]	0.03 [0.37]
	(3)	0.79 [3.25]	0.79 [3.50]	0.81 [3.37]	0.89 [3.92]	0.85 [3.69]	0.06 [0.82]	0.10 [1.39]	0.11 [1.47]	0.10 [1.36]	0.10 [1.32]	0.10 [1.27]
	(4)	0.69 [3.10]	0.81 [3.82]	0.88 [3.95]	0.79 [3.75]	0.83 [3.83]	0.13 [1.70]	0.15 [1.98]	0.15 [1.87]	0.12 [1.53]	0.11 [1.40]	0.10 [1.17]
	(5)	0.50 [2.49]	0.59 [3.37]	0.59 [3.02]	0.67 [3.72]	0.77 [3.99]	0.27 [3.16]	0.30 [3.42]	0.31 [3.57]	0.29 [3.25]	0.24 [2.68]	0.23 [2.57]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PBR Quintiles					PBR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	383	385	384	385	381	36	33	31	33	34	
	(2)	106	106	106	106	106	60	60	59	60	60	
	(3)	76	76	76	76	76	108	108	104	107	106	
	(4)	64	64	64	64	64	228	240	230	235	227	
(5)	59	59	59	59	59	1320	2184	2013	2220	1439		

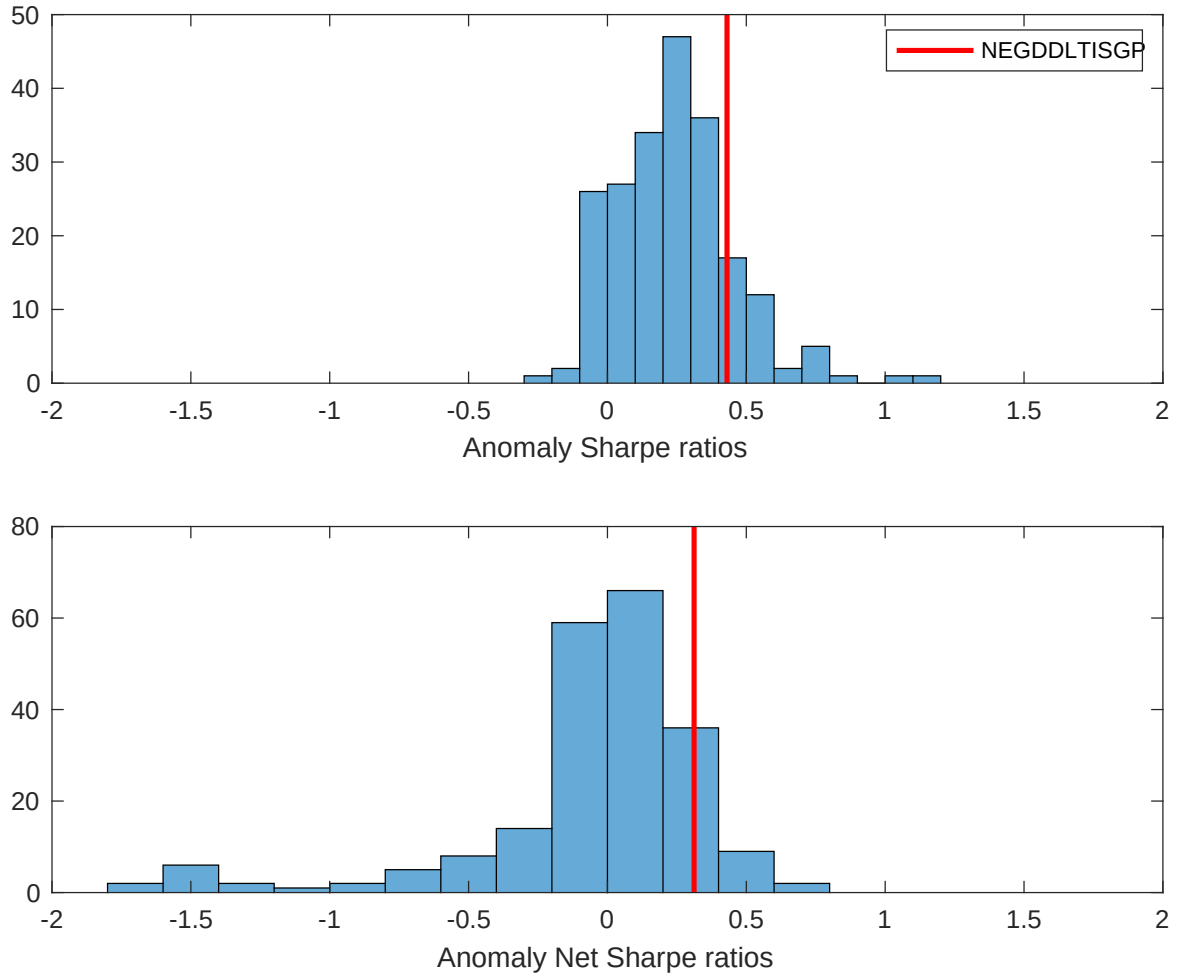


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

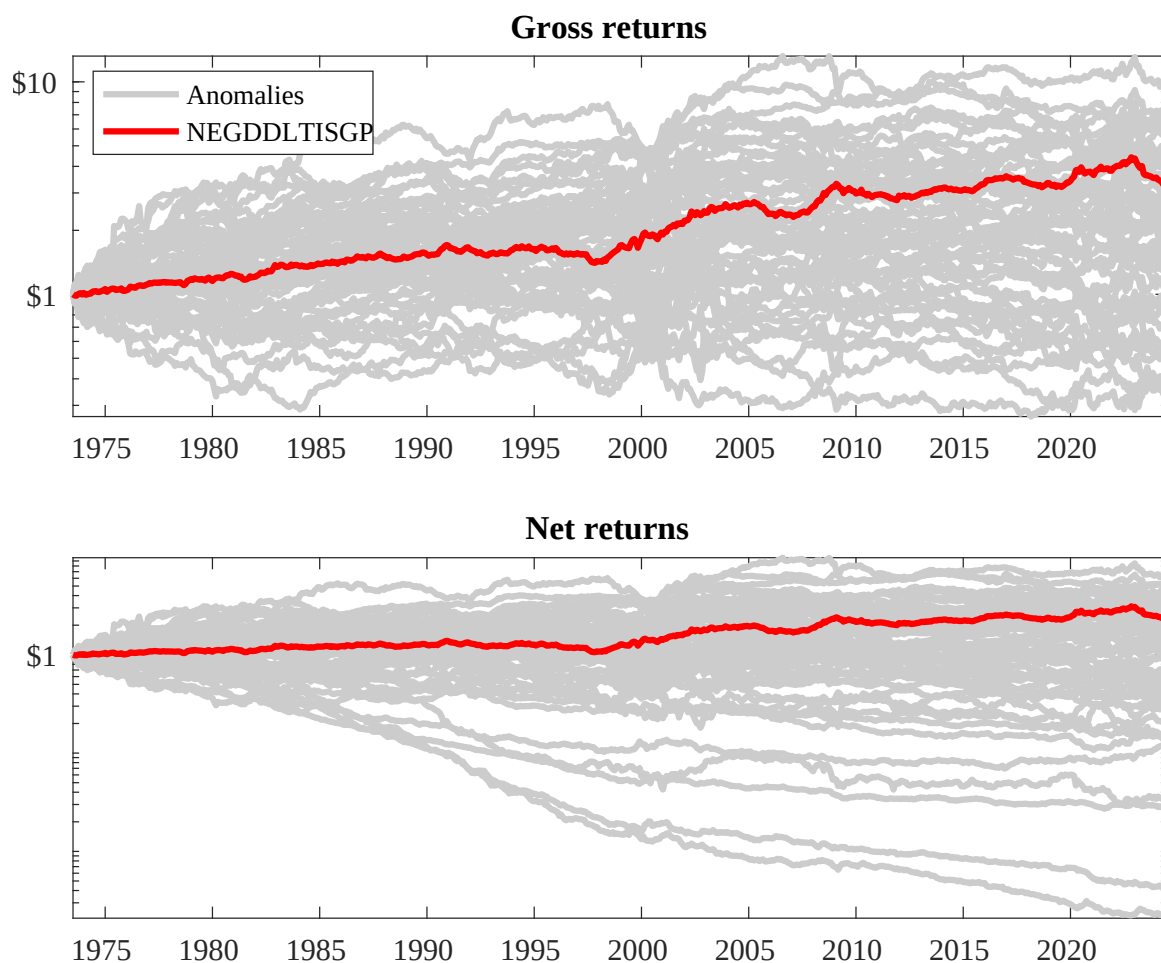
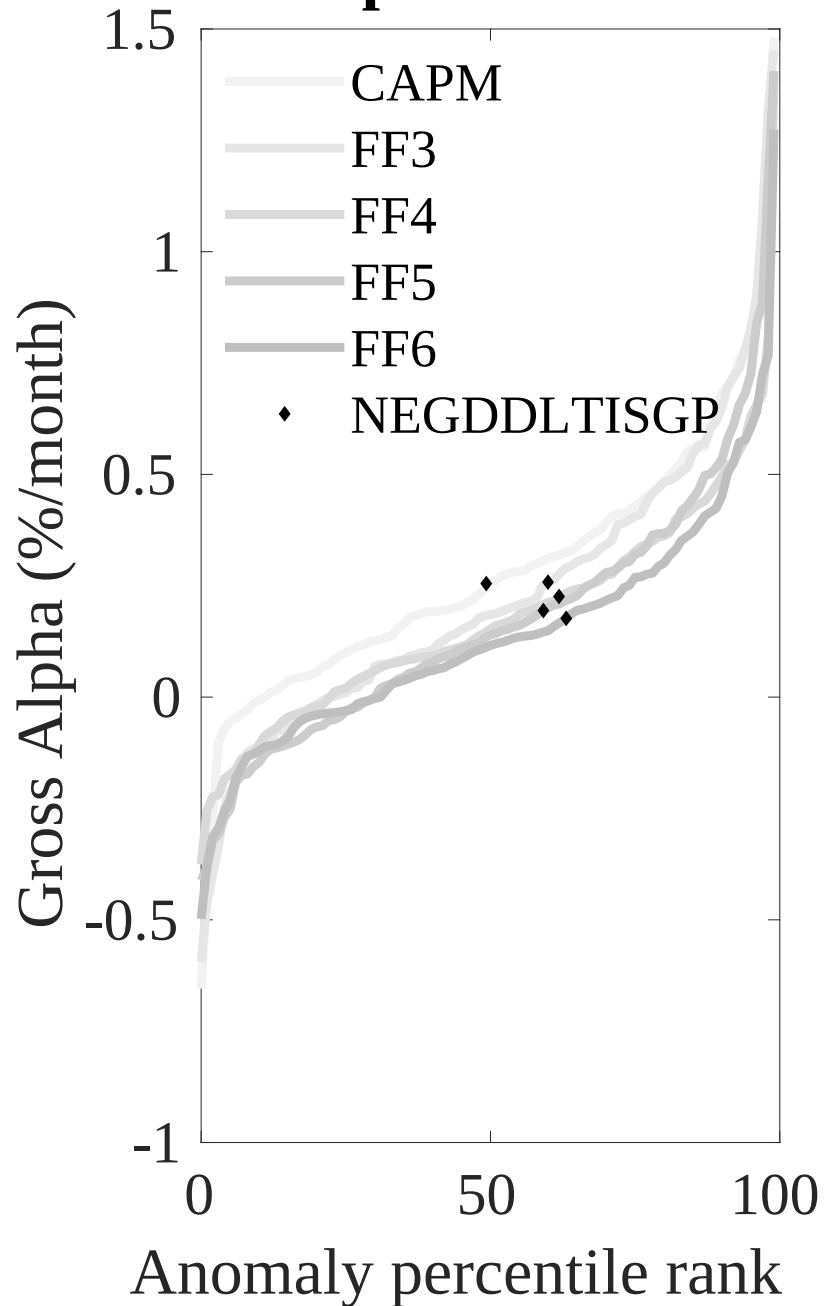


Figure 3: Dollar invested.

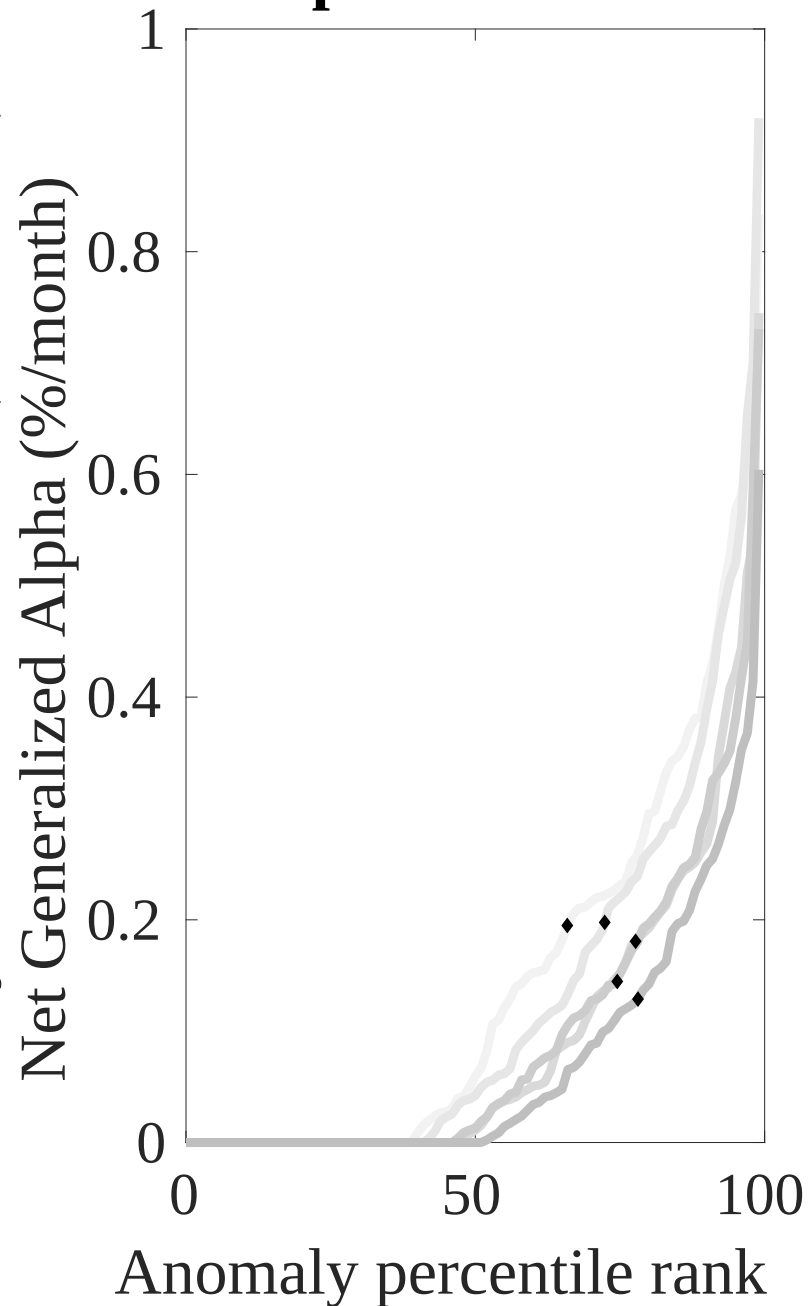
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



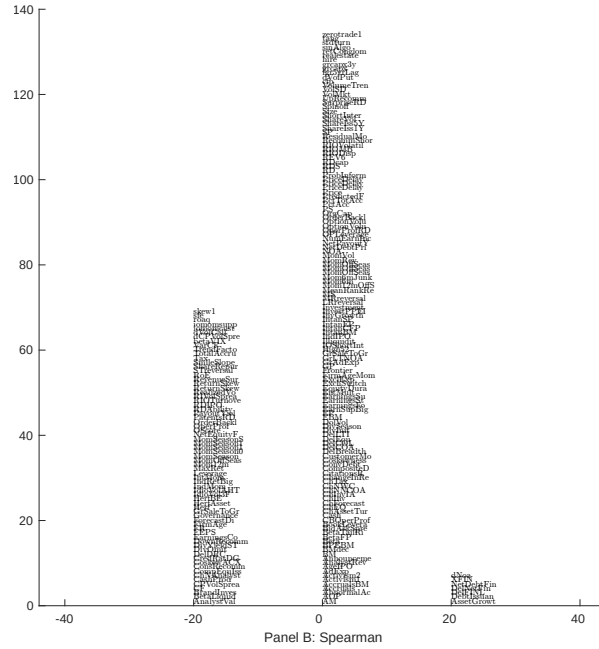
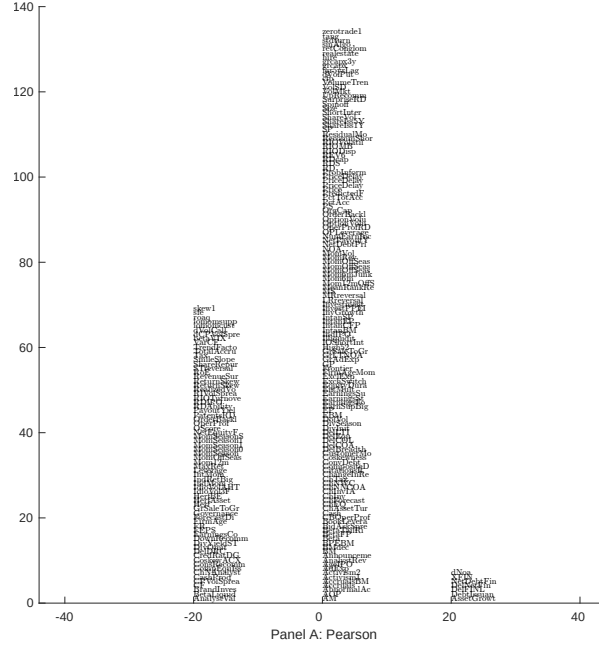


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with PBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

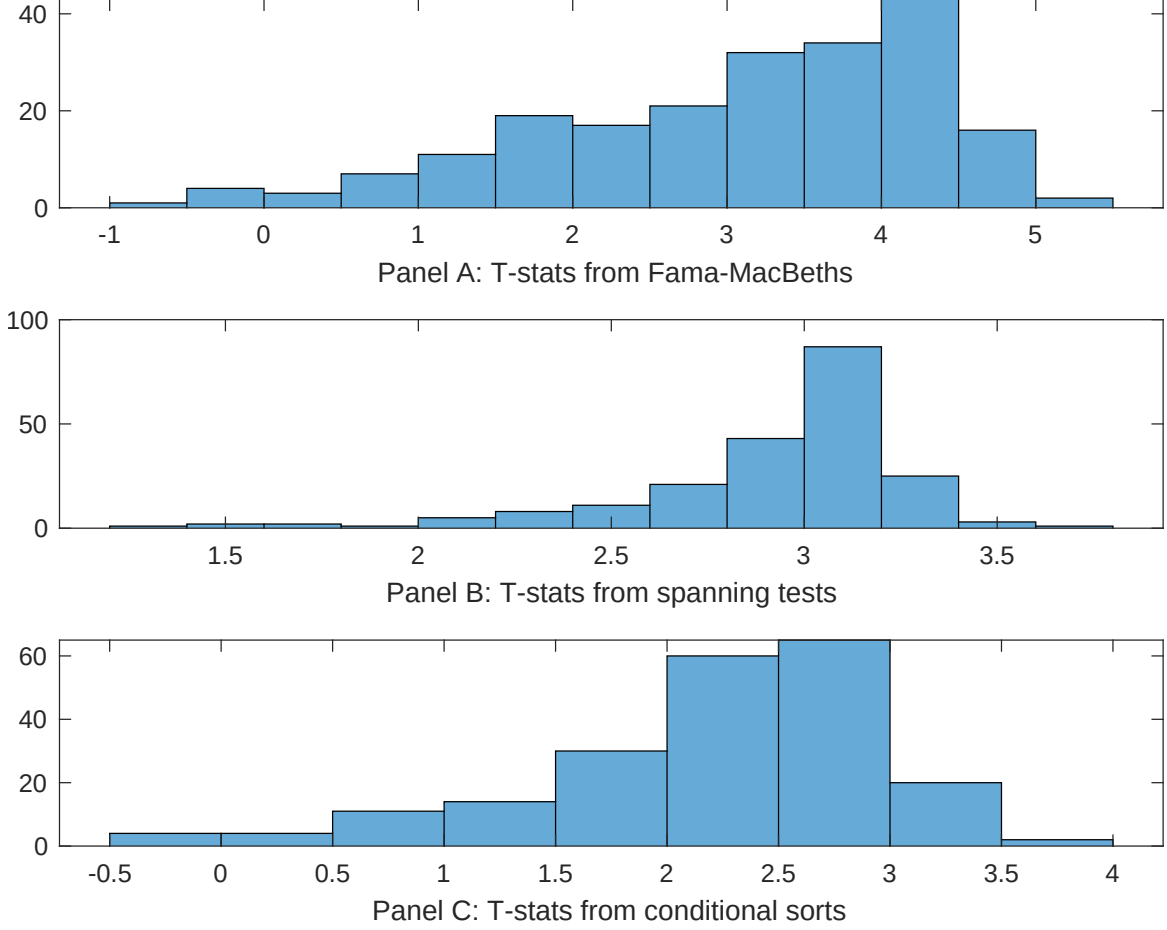


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR}PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PBR}PBR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.44]	0.13 [5.40]	0.14 [5.70]	0.14 [5.32]	0.14 [5.75]	0.14 [5.88]	0.15 [5.71]
PBR	0.23 [0.13]	0.59 [0.33]	0.29 [1.58]	0.82 [3.49]	-0.69 [-0.41]	-0.14 [-0.08]	-0.97 [-0.39]
Anomaly 1	0.18 [9.63]						-0.11 [-2.23]
Anomaly 2		0.21 [9.73]					0.10 [1.52]
Anomaly 3			0.20 [6.56]				0.96 [1.64]
Anomaly 4				0.41 [7.03]			0.38 [0.69]
Anomaly 5					0.15 [10.16]		0.94 [5.21]
Anomaly 6						0.11 [8.77]	0.42 [1.84]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PBR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15 [2.14]	0.16 [2.27]	0.16 [2.23]	0.18 [2.54]	0.16 [2.26]	0.17 [2.44]	0.14 [2.10]
Anomaly 1	24.32 [6.27]						18.89 [3.54]
Anomaly 2		22.46 [5.98]					7.44 [1.46]
Anomaly 3			14.40 [4.13]				11.20 [3.05]
Anomaly 4				7.24 [2.65]			6.55 [2.39]
Anomaly 5					8.54 [2.12]		-3.22 [-0.73]
Anomaly 6						5.21 [1.27]	-1.44 [-0.33]
mkt	-3.90 [-2.47]	-4.42 [-2.80]	-2.49 [-1.49]	-4.60 [-2.85]	-4.42 [-2.73]	-4.47 [-2.76]	-2.59 [-1.58]
smb	2.53 [1.05]	3.04 [1.26]	9.66 [3.62]	5.83 [2.37]	5.18 [2.11]	4.66 [1.88]	6.79 [2.45]
hml	-10.97 [-3.60]	-11.66 [-3.83]	-10.97 [-3.52]	-12.80 [-4.11]	-13.31 [-4.24]	-12.80 [-4.08]	-9.68 [-3.15]
rmw	-1.27 [-0.41]	-1.55 [-0.50]	-7.69 [-2.04]	1.97 [0.62]	1.06 [0.34]	0.94 [0.30]	-7.40 [-1.98]
cma	17.45 [3.65]	19.76 [4.18]	16.19 [3.11]	19.14 [3.60]	19.29 [3.46]	19.26 [2.81]	8.37 [1.22]
umd	0.57 [0.36]	1.23 [0.78]	2.58 [1.62]	1.94 [1.19]	2.39 [1.48]	2.79 [1.73]	-0.12 [-0.08]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	14	13	11	9	9	8	16

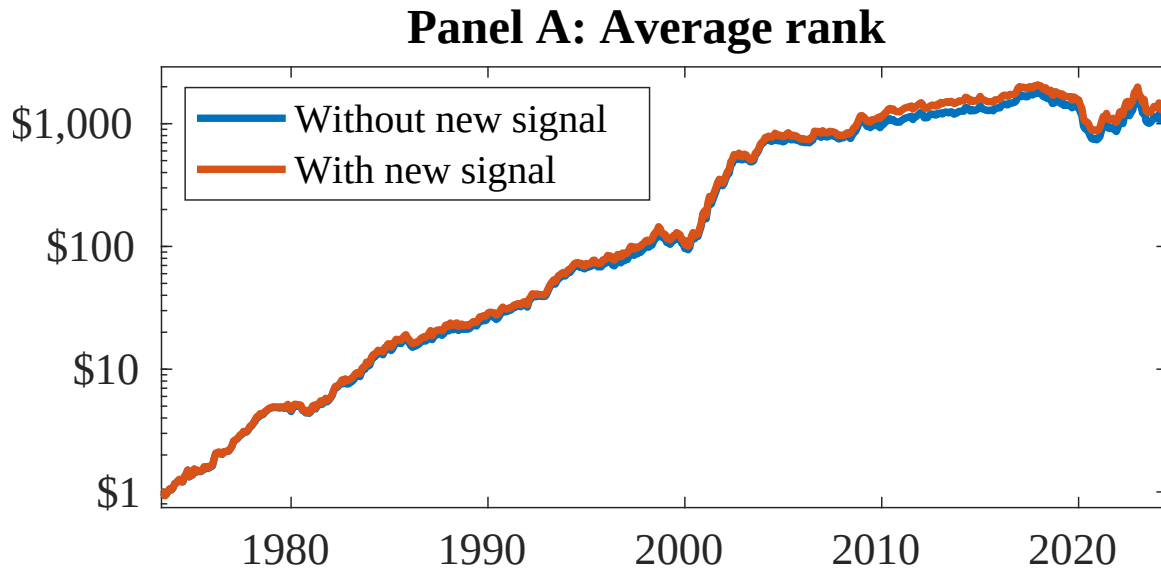


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as PBR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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